

Maurice Karnaugh (k-map)

With the help of k-map method, we find the simplest POS and SOP expression which is known as the minimum expression. The k-map method is used for expression containing 2, 3, 4 and 5 variables. For higher number of variables, there is another method used for simplification called the Quine Mc. Clusky method.

There are following steps used to solve the expression using k-map:

1. First, we find the k-map, as per the number of variables
2. Find the maxterm (POS) or Minterm (SOP) in the given expression
3. Fill cells of k-map for SOP with "1" respective to minterms or Fill cells of k-map for POS with "0" respective to maxterms.
4. Next we create rectangular groups that contain total terms in the power of two, like 1, 2, 4, 8, ... and try to cover as many elements as we can in one group. [Octate (8-members), quad (4-member), pair (2-members), singlet (1-member)]
5. With the help of these groups, we find the product terms and sum them up for the answer or simplified expression.

2-variable map,

$$2^2 = 4$$

Number of cells = 2^n $[\because n > 1]$
 n = number of variables.

		$\overline{B}$ B	
		$\overline{B}$	B
$\overline{A}$	A	0	1
	$\overline{A}$	2	3

SOP

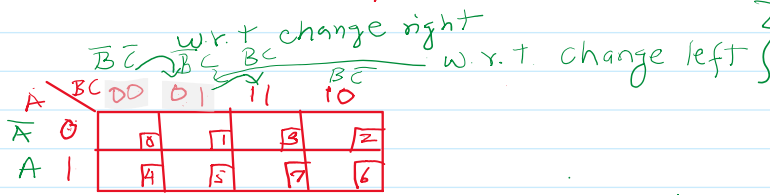
		$\overline{B}$ B	
		$\overline{B}$	B
$\overline{A}$	A	0	1
	$\overline{A}$	2	3

POS

3-variable-map $2^3 = 8$

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3-variable-map $2^3 = 8$



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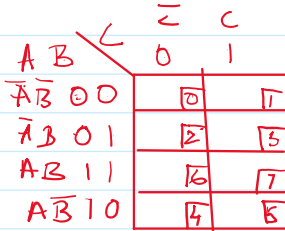
00 → 01 → 11 → 10

SOP

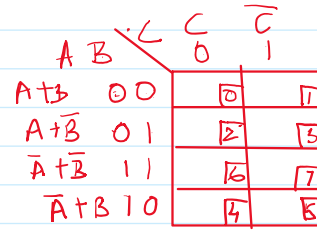
OR



POS



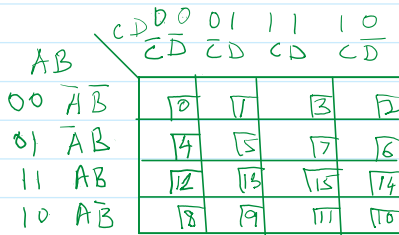
SOP



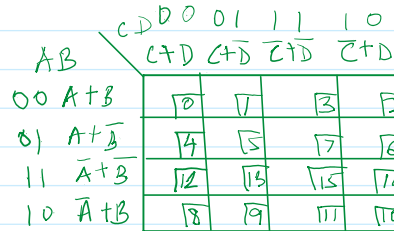
POS.

4-variable map

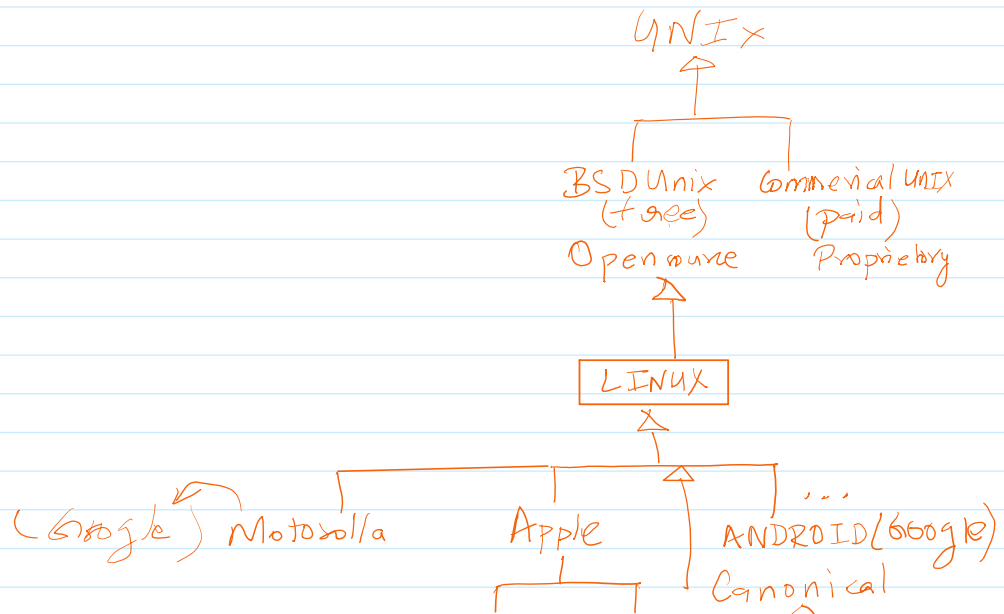
$2^4 = 16$



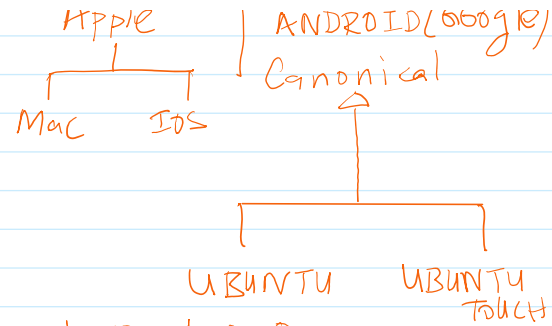
SOP



POS



(Google) Motorola



5-variable map

$\bar{X}$

AB \ CD	00	01	11	10
00 $\bar{A}\bar{B}$	1	1	1	1
01 $\bar{A}B$	1	1	1	1
11 AB	1	1	1	1
10 $A\bar{B}$	1	1	1	1

SOP

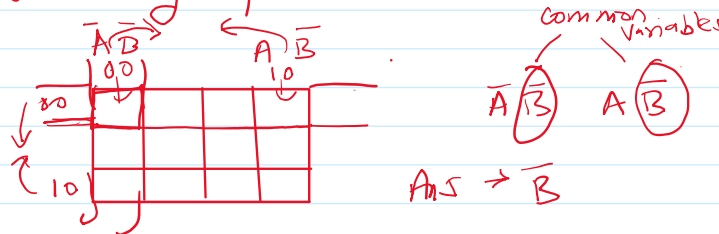
$X=1$

AB \ CD	00	01	11	10
00 $\bar{A}\bar{B}$	1	1	1	1
01 $\bar{A}B$	1	1	1	1
11 AB	1	1	1	1
10 $A\bar{B}$	1	1	1	1

SOP

1	0	1	0	0
16	8	4	2	1
X	A	B	C	D
0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1
1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1

Note: Map can fold horizontally or vertically to form a group.



Rule: 1st look at row-wise
2nd look at column-wise.
and select common variables.

Question 3

(i) Given the Boolean function $F(A,B,C,D) = \sum(0, 1, 2, 3, 4, 6, 9, 11, 13)$.

(a) Reduce the above expression by using 4-variable Karnaugh map, showing the various groups (i.e. octal, quads and pairs). [4]

(b) Draw the logic gate diagram for the reduced expression. Assume that the variables and their complements are available as inputs. [1]

(a)

4 variable SOP

AB \ CD	00	01	11	10
00 $\bar{A}\bar{B}$	1	1	1	1
01 $\bar{A}B$	1	1	1	1
11 AB	1	1	1	1
10 $A\bar{B}$	1	1	1	1

quad $m_1(0,1,3,2) = \bar{A}\bar{B}$
quad $m_2(0,2,4,6) = \bar{A}\bar{D}$
quad $m_3(1,3,9,11) = \bar{B}D$
pair $m_4(13,9) = A\bar{C}D$

01 $\bar{A}\bar{B}$	1	4	5	7	8
→ 11 $A\bar{B}$	12	1	5	15	14
→ 10 $A\bar{B}$	18	1	9	1	11

SOP

group $m_2 (0, 2, 4, 6) = \bar{A}\bar{D}$
 group $m_3 (1, 3, 9, 11) = \bar{B}D$
 pair $m_4 (13, 9) = A\bar{C}D$

$$F(A, B, C, D) = \bar{A}\bar{B} + \bar{A}\bar{D} + \bar{B}D + A\bar{C}D$$

